

Section X: Cybersecurity & Cyber-Resilience Strategy for Sovereign Cloud/Edge Computing

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Security Framework with Separated Security Verticals

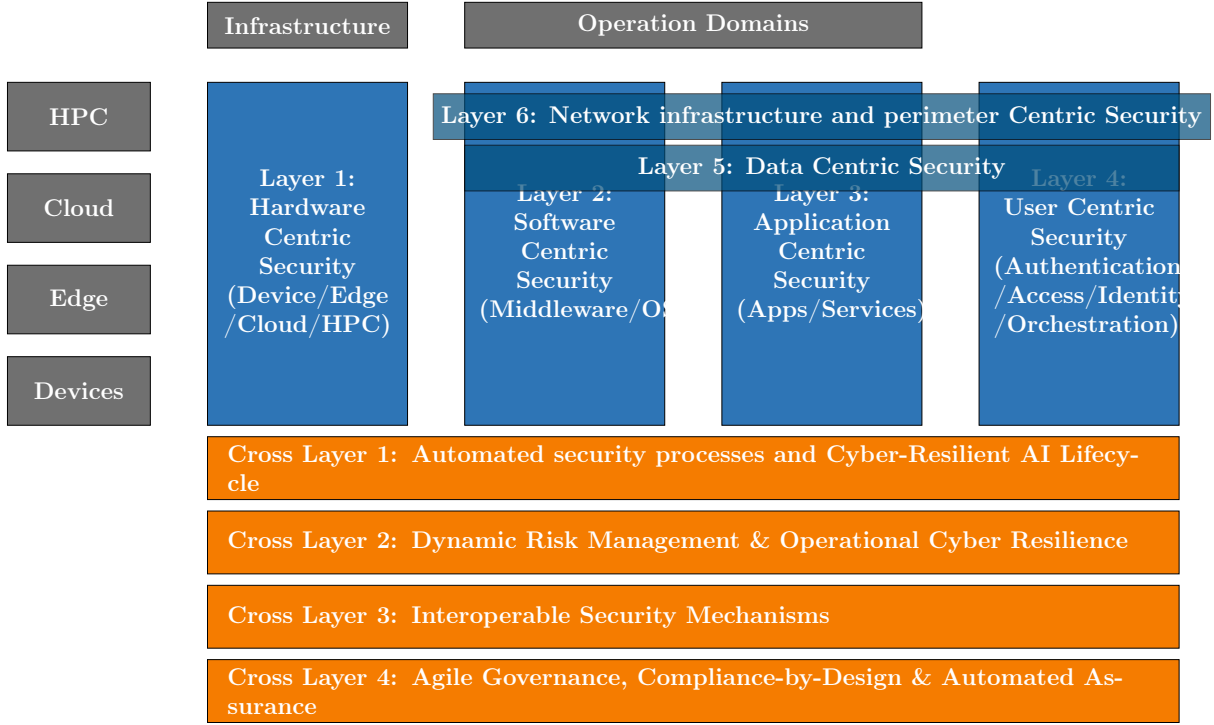


Figure 1: Security Framework with separated security verticals in dedicated boxes.

Topic: Cybersecurity for the Cognitive Computing Continuum

Destinations:

- **Primary:** 4. A secure, sovereign European computing continuum infrastructure.
- **Secondary:** 6. European leadership in emerging disruptive computing paradigms.

1 Background & Driving Factors

- **Decentralization Risk:** The paradigm shift from the centralized Cloud to a distributed Cloud-Edge-IoT continuum radically expands the attack surface. This creates environments that are often “too many, too diverse, and too dispersed” to secure with traditional perimeter defenses.
- **Physical Exposure:** Edge nodes and far-edge IoT devices frequently operate in the field. Unlike hyperscaler data centers, they lack strict physical and environmental security controls, making them highly vulnerable to tampering.
- **Regulatory Pressure:** There is an immediate, complex need to operationalize the NIS2 Directive [1], the Cyber Resilience Act (CRA) [2], and the AI Act [3] across a highly fragmented technological ecosystem to ensure compliance and market access.
- **Digital Sovereignty:** Europe must urgently reduce its systemic reliance on non-EU technology providers. Doing so will mitigate extra-territorial data access risks

and avoid geopolitical vendor lock-in, ensuring independent control over critical infrastructure.

2 Europe in Relation to the State of the Art

Comparison vs. Hyperscalers:

- **EU Strengths:** Europe leads globally in Trust, Privacy-by-Design, proactive regulation (GDPR, Data Act, AI Act), and rigorous, standardized certification frameworks (e.g., EUCS [4]).
- **EU Gaps:** The European market suffers from a lack of unified, hyper-scale security orchestration. Furthermore, there is a critical dependency on non-EU hardware (secure silicon) and highly fragmented threat intelligence sharing among member states.

Strategic Goal: To definitively transition the European computing continuum from a state of “reactive, siloed security” to an advanced architecture defined by “autonomous, federated, and certified Security-by-Design.”

3 Strategic Preamble

- **Context:** This section defines the architectural guardrails required to secure the Cloud-Edge-IoT Continuum. It moves beyond traditional “perimeter security” toward a model of Cyber-Resilience—where systems are engineered to proactively anticipate, withstand, recover from, and adapt to attacks while maintaining strict Digital Sovereignty.
- **Approach:** To provide actionable, risk-aware Research & Innovation (R&I) priorities, the strategy is organized into a two-part framework:
 - **Capability Profiles (Qualitative):** A deep-dive 7-dimensional analysis defining exactly what needs to be built.
 - **Progression Matrix (Quantitative & Strategic):** A structured timeline tracking the evolution of capabilities through their Operational Deployment Phases and their trajectory toward Strategic Sovereignty.

4 The Core Capability Dimensions (Architectural Framework)

The roadmap is structured across a cohesive architectural framework that spans four continuum environments (HPC, Cloud, Edge, Devices). The capabilities are organized into Six Security Layers (The Vertical Stack) and Four Transversal Cross-Layers (The Horizontal Enablers).

Part A: The Security Layers (Vertical Stack)

These layers define the specific technical security controls implemented across the computing infrastructure.

Layer 1: Hardware-Centric Security

1.1 Hardware Roots of Trust: Cryptographically verifiable supply chain trust (e.g., TPM attestation, secure enclaves) ensures physical device integrity. Success depends on driving European open-hardware designs (e.g., RISC-V) toward **EUCC 'High' assurance level certification** [5] to mathematically guarantee resistance against state-level physical tampering and side-channel attacks at the vulnerable far-edge.

Layer 2: Software-Centric Security (Middleware/OS)

2.1 OS & Middleware Hardening: Securing hypervisors, container runtimes, and distributed operating systems is critical for isolation. This involves hardening KVM/QEMU and supporting sovereign, lightweight Linux distributions optimized for edge resource constraints, ensuring a reduced attack surface for the cognitive continuum.

Layer 3: Application-Centric Security (Apps/Services)

3.1 Workload Security: Securing containerized applications, microservices, and serverless functions executing dynamically across multi-provider nodes. This is achieved through hardened service meshes and sidecar proxy architectures that provide mutual TLS and granular traffic inspection without impacting performance.

Layer 4: User & Orchestration-Centric Security

4.1 Zero-Trust Orchestration: Seamless policy enforcement across a multi-provider computing continuum requires robust Policy Enforcement Point (PEP) and Policy Decision Point (PDP) architectures. This enables continuous authorization and dynamic access controls that adapt to the context of the request.

4.2 Federated Identity & Access Management (IAM): Cross-domain identity federation and verifiable credentials for machines, users, and automated workloads. Implementing **Self-Sovereign Identity (SSI)** principles allows for cryptographically secure, decentralized authentication across heterogeneous cloud-edge domains.

Layer 5: Data-Centric Security

5.1 End-to-end Confidential Computing: Data-in-use security through Trusted Execution Environments (TEEs) spanning the entire Cloud-to-IoT data path. Leveraging TEE abstraction layers and remote attestation ensures that sensitive European data remains encrypted even during processing in non-trusted environments.

5.2 Post-Quantum & Cryptographic Agility: PQC standardization and hybrid crypto deployment are necessary to secure long-lived data. Priority is placed on **PQC migration for resource-constrained devices**, ensuring that edge infrastructure is resilient against "Harvest Now, Decrypt Later" quantum threats.

Layer 6: Network Infrastructure & Perimeter-Centric Security

6.1 Telco-cloud & O-RAN Security: Hardening distributed 5G/6G cores and RAN components must align with the **Smart Networks and Services Joint Undertaking (SNS JU) SRIA** [7]. This ensures sovereign control over network slice isolation and virtualized RAN components.

6.2 Cross-Sector Physical Resilience & Safety: Adapting strict network controls and IT/OT gateway security for high-availability edge environments (e.g., Smart Grids). This guarantees that cyber-breaches in the digital layer do not cascade into physical safety hazards or kinetic damage in critical infrastructure.

Part B: The Transversal Security Verticals (Cross-Layers)

These four cross-layers represent continuous, automated processes and governance structures that intersect every layer of the vertical stack.

Cross-Layer 1: Automated Security Processes & Cyber-Resilient AI Lifecycle

CL1.1 Continuous DevSecOps (Shift-Left): Embedding AI-driven code analysis and automated vulnerability remediation at the earliest stages of the software lifecycle.

CL1.2 Model & Pipeline Integrity (DataBOMs): Securing AI training pipelines to prevent data poisoning and unauthorized model manipulation.

CL1.3 Runtime AI Defense: Defending AI/ML infrastructure against real-world adversarial attacks (e.g., prompt injection, model evasion/theft) at the edge.

Cross-Layer 2: Dynamic Risk Management & Operational Cyber Resilience

CL2.1 Automated SOC Operations & Collaborative Threat Intelligence: Continuous monitoring, SOC federation across Member States, and automated incident containment, underpinned by legally protected, cross-border threat intelligence sharing ecosystems designed to overcome institutional trust barriers and incentivize active collaboration [9].

CL2.2 Systemic & Cascading Risk Mitigation: Mapping the evolving threat landscape and using impact-likelihood scoring to dynamically sever links and prevent cross-domain systemic failures.

Cross-Layer 3: Interoperable Security Mechanisms & Open-Source Verifiability

CL3.1 Heterogeneous Interoperability & Simpl Integration: Secure communication and data exchange across disparate systems, edge nodes, and proprietary domains. This includes mandatory alignment with and integration into the Simpl open-source smart middleware [6] to ensure secure, federated access across Common European Data Spaces.

CL3.2 Minimal Interoperability Mechanisms (MIMs) & Open Verifiability: Establishing standardized, baseline security and privacy controls to ensure trust without creating vendor lock-in. This relies heavily on embracing open-source software and open hardware architectures as fundamental prerequisites for transparency, cryptographic verifiability, and the elimination of proprietary black-box vulnerabilities [6].

Cross-Layer 4: Agile Governance, Compliance-by-Design & Automated Assurance

CL4.1 Policy-as-Code & Compliance-by-Design: Technical implementation of legal requirements (OSCAL) into machine-readable rules that pre-validate workloads.

CL4.2 Continuous Automated Auditing & Certification Lifecycle: Replacing “point-in-time” PDF certificates with real-time, telemetry-driven certification tracking for both cloud services (EUCS) [4] and physical ICT edge products (EUCC) [5]. This ensures that edge nodes autonomously maintain compliance by integrating mandatory vulnerability handling and cryptographic patch verification directly into their operational lifecycle, strictly aligning with EUCC maintenance and assurance rules [5].

CL4.3 Regulatory Digital Twins: Virtualized replicas of critical infrastructure specifically to stress-test regulatory compliance against evolving laws before applying changes to the live physical continuum.

5 The 4-Filter Prioritization Framework

To ensure the roadmap provides actionable guidance for European R&I funding (Horizon Europe) and strategic investment, capabilities are prioritized using a structured 4-filter logic. This framework identifies the “Critical Path” items that require immediate resource allocation in Phase 1 (0-5 Yrs).

1. Regulatory Urgency (The “Must-Do” Filter):

- **Question:** Is this capability strictly required to comply with imminent EU law (NIS2 [1], Cyber Resilience Act [2], AI Act [3])?
- **Logic:** If failure to achieve this by 2026 results in market exclusion or systemic non-compliance, it is designated as **Priority #1**.

2. Foundational Dependency (The “Blocker” Filter):

- **Question:** Do subsequent security layers depend on this capability existing first?
- **Logic:** Foundational technologies (e.g., Layer 1 Hardware Trust) outrank high-level software orchestration. You cannot secure the user (Layer 4) if the hardware (Layer 1) cannot provide a verifiable anchor.

3. The Sovereignty Cliff (The “Extinction” Filter):

- **Question:** Does a lack of this tech pose an existential threat to EU sovereignty or systemic resilience?
- **Logic:** This prioritizes areas where the EU currently has 100% foreign dependency or where a technical breakthrough (e.g., *Q-Day* Quantum threat) would render existing defenses obsolete.

4. Market Feasibility & Ecosystem (The “Simpl/SME” Filter):

- **Question:** Does this align with existing EU heavy-investments (e.g., Simpl [6]) and is it adoptable by SMEs?
- **Logic:** Priority is given to capabilities that act as force-multipliers for existing infrastructure, ensuring economic viability.

Executive Priorities: The Critical Path (Phase 1)

Applying the 4-Filter Framework, the following areas are identified as immediate strategic priorities for the 2025-2030 funding cycle:

- **Priority 1:** Hardware Roots of Trust & EUCC 'High' Certification (Filters 1, 2, 3).
- **Priority 2:** Post-Quantum Cryptographic Agility for Edge/IoT (Filter 3).
- **Priority 3:** Security Interoperability via Simpl Middleware (Filter 4).

6 The Anatomy of a Roadmap Priorities: A 7-Dimensional Analysis

To ensure the roadmap is actionable, realistic, and aligned with European strategic autonomy, every priority is analyzed holistically through a strict 7-point structural template:

1. **Baseline & Target Objective (The Trajectory) - Q: Where do we stand today, and what exactly does success look like?**

Description: Establishes the starting point of the priority today and defines the ultimate maturity state required by the end of the 15-year roadmap.

- **Baseline (Where we are):** Grounds the roadmap in reality. It captures the state of the art, current Operational Deployment Phase (ODP), Strategic Sovereignty Level (SSL), industrial capacity, reliance on foreign vendors, and the technical or economic barriers preventing immediate adoption.
- **Target (Where we need to be):** Sets the baseline for EU-funded projects, defining the target ODP and SSL that must be achieved by the end of the roadmap.

2. **Primary Drivers (The “Push” Factors) - Q: Why must we prioritize building this capability right now? SQ: Why are we doing this?**

Description: Outlines the external pressures—both from the evolving threat landscape and mandatory EU regulations—that make this technological evolution urgent and unavoidable.

- **Threat Drivers:** The rapid advancement of adversarial capabilities (e.g., offensive AI, quantum decryption, supply chain targeting).
- **Regulatory Drivers:** Imminent EU compliance mandates that require technical operationalization (e.g., NIS2, Cyber Resilience Act, AI Act).

3. **High Impact Activities - Q: Which specific actions provide the greatest leap in European sovereignty and security? SQ: Which should be the key initiatives?**

Description: Highlights the specific, prioritized technical interventions within this broader capability group that will yield the highest strategic return on investment.

4. **Catalysts & Enablers (The “Pull” Factors & EU Support) - Q: What specific EU funding, standards, and ecosystem support do we need? SQ: How do we get there & accelerate?**

Description: Identifies the concrete mechanisms, funding vehicles, and regulatory standardizations required from the EU to accelerate market adoption and push the industry from the baseline to the target state.

- **R&I & Funding Alignment:** Required Horizon Europe funding topics and targeted innovation instruments. Investment priorities must be explicitly synchronized with the European Cybersecurity Competence Centre (ECCC) Strategic Agenda [8] to ensure alignment with pan-European goals (e.g., PQC transition infrastructure, AI-driven security tools).
- **Regulatory & Standardization:** Alignment with standard-setting bodies (CEN/CENELEC) and certification frameworks (e.g., EUCS, EUCC).
- **Ecosystem:** Support for ecosystem building, industrial partnerships, and capacity-building initiatives (e.g., federated Cyber Ranges) to upskill the workforce and ensure advanced security solutions are economically viable for SMEs.

5. **Disruptors & Systemic Risks (The Roadblocks) - Q: What structural vulnerabilities, market failures, or adversary breakthroughs could derail us? SQ: What can break the path?**

Description: Acknowledges that innovation is not linear by identifying severe technical and architectural risks that could actively derail deployment. It places primary emphasis on “Technical Cliffs”—moments where defensive capabilities are rendered obsolete by adversarial breakthroughs before countermeasures are fully deployed.

- **Cryptographic Obsolescence & Migration Bottlenecks (The Primary Technical Risk):**
 - *The “Q-Day” Cliff:* The risk that Cryptographically Relevant Quantum Computers (CRQC) become operational before the EU completes its migration to Post-Quantum Cryptography (PQC). This creates a “Harvest Now, Decrypt Later” vulnerability.
 - *Migration Complexity:* The systemic failure to inventory and replace deep-rooted Public Key Infrastructure (PKI) in legacy OT/IoT environments.
- **Adversarial Breakthroughs:** The risk of threat actors weaponizing emerging technologies (e.g., AI-driven exploit generation, polymorphic malware) faster than defensive R&I can mitigate them.
- **Contextual Risks (Non-Technical / Optional):** Geopolitics & Supply Chain dependencies, Market & Tech Risks (monopolization, fragmented standards).

6. **Failsafe Mechanisms & Resilience (The Safety Nets) - Q: If this technology is compromised, how does the system survive and gracefully degrade? SQ: How do we prevent failure?**

Description: The technical core of “Cyber-Resilience.” It defines the engineering fallbacks required if a primary capability fails, is breached, or is disrupted by the systemic risks identified above.

What it entails: Designing for degraded modes of operation, architectural redundancies, autonomous isolation protocols, and implementation diversity to avoid vulnerable technological monocultures.

7. **Transversal Meta-Layer (The Policy Alignment) - Q: How does mastering this capability support Europe’s goals of sovereignty, sustainability, and trust? SQ: The broader policy context?**

Description: Elevates the technical capability back to the overarching political and societal goals of the European Union.

- **Sovereignty & Autonomy:** Strengthening European strategic independence, reducing reliance on non-EU vendors.
- **Trust & Ethics:** Fostering deep Societal Trust by guaranteeing data privacy (Privacy-by-Design), algorithmic fairness, and transparency.
- **Sustainable Security (Green IT):** Minimizing the environmental impact of cyber defense (e.g., energy-efficient cryptographic algorithms, crypto-agility to reduce e-waste).

7 Progression Matrix: Sequence of Events (X-Axis)

Instead of a strictly chronological timeline, the roadmap maps the sequence of critical events required to unlock maturity. This ensures that foundational capabilities are built before advanced autonomous systems.

- **Phase 1: Foundation & Compliance (Immediate Implementation):** Focuses on state-of-practice deployments, bridging immediate capability gaps, and addressing urgent regulatory obligations (NIS2, CRA).
- **Phase 2: Federation & Standardization (Scaling & Harmonization):** Focuses on deploying emerging technologies, establishing harmonized governance frameworks, and managing critical transition periods.
- **Phase 3: Autonomy & Leadership (Sovereign Innovation):** Focuses on highly speculative but structured foresight, including defense against quantum-capable adversaries and the achievement of fully sovereign, autonomous cloud-edge ecosystems.

Matrix Cell Content Definition (Measuring Progression)

Rather than relying on subjective Technology Readiness Levels (TRL) or Manufacturing Readiness Levels (MRL) for a highly fragmented ecosystem, each cell in the progression matrix tracks maturity using objective strategic market and policy indicators:

- **Operational Deployment Phase (ODP):** Tracks the market reality and integration status (e.g., *Applied R&D* → *Fragmented Piloting* → *Standardized Integration* → *Ubiquitous Operation*).
- **Strategic Sovereignty Level (SSL):** Tracks European strategic autonomy and reliance on foreign vendors (e.g., *High Foreign Dependency* → *Emerging EU Ecosystem* → *Full Sovereign Autonomy*).

- **Operational State:** A concise snapshot characterizing the reality on the ground during this phase.
- **Critical Enabler:** The single most critical regulation, standard, or European R&I breakthrough unlocking the transition to the next phase.

8 Capability Profiles & Progression Matrices

Progression Matrix: [Insert Capability Name]

Metric	Phase 1: Foundation & Compliance (0-5 Yrs)	Phase 2: Federation & Standardization (5-10 Yrs)	Phase 3: Autonomy & Leadership (10-15 Yrs)
Deployment Phase (ODP)	Fragmented Piloting	Standardized Integration	Ubiquitous Operation
Sovereignty Level (SSL)	High Foreign Dependency	Emerging EU Ecosystem	Full Sovereign Autonomy
Operational State	[Cloud vs Edge snapshot here]	[Cloud vs Edge snapshot here]	[Cloud vs Edge snapshot here]
Critical Enabler	[Regulation/Standard here]	[Standard/Funding here]	[Autonomy milestone here]

9 Master Population Prompt for Roadmap Generation

Context: You are an expert EU Cybersecurity Policy Strategist. Your task is to populate the content for a Horizon Europe roadmap document based on the provided Master LaTeX structure.

Phase 1: Populate Section 4 (Architectural Summaries)

Using a `description` environment, provide a 2-3 sentence strategic summary for every Layer and Cross-Layer.

- **Layers 1–6:** Connect the technical component to its role in the continuum.
- **Cross-Layers 1–4:** Emphasize automation and governance.
- **Mandatory Hooks:** Reference **SNS JU SRIA** in 6.1, **Simpl Middleware** and **Common European Data Spaces** in CL3.1, and **EUCS/EUCC** in CL4.2.

Phase 2: Priority Selection & 7-Dimensional Analysis

Apply the **4-Filter Prioritization Framework** (Regulatory Urgency, Foundational Dependency, Sovereignty Cliff, and Market Feasibility) to justify the selection of the Top 3 Priorities:

1. **Priority 1:** Hardware Roots of Trust & EUCC 'High' Certification.
2. **Priority 2:** Post-Quantum Cryptographic (PQC) Agility for Edge/IoT.

3. **Priority 3:** Security Interoperability via Simpl Middleware.

For each priority, generate a full analysis using the **7-Dimensional Template** (Baseline/Target, Drivers, High Impact Activities, Catalysts/ECCC, Disruptors/Q-Day, Fail-safe, and Policy Meta-Layer).

Phase 3: Progression Matrices

Generate a LaTeX `tabular` matrix for each priority using the **ODP** (Operational Deployment Phase) and **SSL** (Strategic Sovereignty Level) metrics across the three time horizons.

- **Phase 1 (0-5 Yrs):** Target NIS2/CRA compliance and bridging foreign dependency.
- **Phase 2 (5-10 Yrs):** Focus on standardized federation and EU-wide certification.
- **Phase 3 (10-15 Yrs):** Achieve full sovereign autonomy and quantum-native resilience.

Tone & Style: Evaluator-grade, formal, and strictly aligned with the *European Industrial Technology Roadmap for the Next-Generation Cloud-Edge*.

10 The Core Capability Dimensions (Architectural Framework)

The roadmap is structured across a cohesive architectural framework that spans four continuum environments (HPC, Cloud, Edge, Devices).

Part A: The Security Layers (Vertical Stack)

Layer 1.1 Hardware Roots of Trust: Cryptographically verifiable supply chain trust ensures physical device integrity from fabrication to deployment. Success depends on driving European open-hardware designs (e.g., RISC-V) toward **EUCC 'High' assurance** certification [5] to guarantee resistance against state-level physical tampering at the vulnerable far-edge.

Layer 2.1 OS & Middleware Hardening: Securing hypervisors and distributed operating systems across the continuum. This involves hardening KVM/QEMU and supporting sovereign, lightweight Linux distributions optimized for edge resource constraints, ensuring a minimal attack surface for cognitive workloads.

Layer 3.1 Workload Security: Securing containerized applications and serverless functions executing dynamically across multi-provider nodes. Implementation of hardened service meshes and sidecar proxy architectures provides mutual TLS and granular traffic inspection without impacting performance.

Layer 4.1 Zero-Trust Orchestration: Seamless policy enforcement across a multi-provider computing continuum requiring robust PEP/PDP architectures. This enables continuous authorization and dynamic access controls that adapt to the specific context of the request.

Layer 4.2 Federated Identity & Access Management (IAM): Cross-domain identity federation using Self-Sovereign Identity (SSI) principles. This allows for cryptographically secure, decentralized authentication for machines, users, and automated workloads across heterogeneous cloud-edge domains.

Layer 5.1 End-to-end Confidential Computing: Data-in-use security through Trusted Execution Environments (TEEs) spanning the entire Cloud-to-IoT data path. Leveraging TEE abstraction layers ensures that sensitive data remains encrypted even during processing in non-trusted environments.

Layer 5.2 Post-Quantum & Cryptographic Agility: PQC standardization and hybrid crypto deployment to secure long-lived data. Priority is placed on PQC migration for resource-constrained devices to neutralize "Harvest Now, Decrypt Later" quantum threats.

Layer 6.1 Telco-cloud & O-RAN Security: Hardening distributed 5G/6G cores and RAN components in strict alignment with the **Smart Networks and Services Joint Undertaking (SNS JU) SRIA** [7]. This ensures sovereign control over network slice isolation and virtualized RAN components.

Layer 6.2 Cross-Sector Physical Resilience & Safety: Adapting strict network controls and IT/OT gateway security for high-availability edge environments (e.g., Smart Grids). This prevents digital breaches from cascading into physical safety hazards or kinetic damage.

Part B: The Transversal Security Verticals (Cross-Layers)

CL1.1 Continuous DevSecOps (Shift-Left): Embedding AI-driven code analysis and automated vulnerability remediation at the earliest stages of the software lifecycle to ensure secure-by-design software delivery.

CL1.2 Model & Pipeline Integrity (DataBOMs): Securing AI training pipelines to prevent data poisoning and unauthorized model manipulation, ensuring the reliability of cognitive decisions at the edge.

CL1.3 Runtime AI Defense: Defending AI/ML infrastructure against real-world adversarial attacks, such as prompt injection and model evasion, specifically for models deployed in decentralized edge nodes.

CL2.1 Automated SOC Operations & Collaborative Threat Intelligence: Continuous monitoring and SOC federation across Member States. This utilizes legally protected, cross-border threat intelligence platforms to overcome institutional trust barriers [9].

CL2.2 Systemic & Cascading Risk Mitigation: Mapping the evolving threat landscape and using impact-likelihood scoring to dynamically sever links and prevent cross-domain systemic infrastructure failures.

CL3.1 Heterogeneous Interoperability & Simpl Integration: Secure communication across disparate systems via mandatory integration into the **Simpl** open-source smart middleware [6] to unlock **Common European Data Spaces**.

CL3.2 MIMs & Open Verifiability: Establishing Minimal Interoperability Mechanisms (MIMs) to ensure trust without vendor lock-in, relying on open-source hardware and software as fundamental prerequisites for transparency [6].

CL4.1 Policy-as-Code & Compliance-by-Design: Technical implementation of legal requirements (OSCAL) into machine-readable rules that pre-validate workloads before they are deployed to the continuum.

CL4.2 Continuous Automated Certification: Replacing point-in-time audits with real-time certification tracking for both cloud services (**EUCS**) and physical edge products (**EUCC**), ensuring compliance throughout the operational lifecycle [5].

CL4.3 Regulatory Digital Twins: Virtualized replicas of critical infrastructure used to stress-test regulatory compliance against evolving laws (e.g., Data Act, AI Act) without disrupting live operations.

11 Capability Profiles & Progression Matrices

Priority 1: Hardware Roots of Trust & EUCC 'High' Certification

1. **Baseline & Target Objective:** *Baseline:* High foreign dependency on proprietary Trusted Platform Modules (TPMs) and secure elements. Far-edge devices lack standardized hardware anchors. *Target:* Ubiquitous deployment of EU-designed, open-source (RISC-V) secure enclaves certified at **EUCC 'High' assurance**, providing a mathematically verifiable foundation for the entire continuum.
2. **Primary Drivers:** *Threat:* State-level physical tampering and side-channel attacks on decentralized edge nodes. *Regulatory:* Mandatory hardware security requirements under the **Cyber Resilience Act (CRA)** and the need for certified anchors for **NIS2** critical infrastructure.
3. **High Impact Activities:** Development of sovereign European Secure Elements based on RISC-V; creation of automated hardware-level remote attestation protocols for large-scale edge fleets.
4. **Catalysts & Enablers:** **EU Chips Act** pilot lines for secure silicon; **ECCC Strategic Agenda** funding for open-hardware security; alignment with CEN/CENELEC hardware security standards.
5. **Disruptors & Systemic Risks:** Supply chain bottlenecks for EU-based fabrication; breakthroughs in physical side-channel analysis that bypass current cryptographic enclaves.
6. **Failsafe Mechanisms & Resilience:** Implementation of "defense-in-depth" where software-based isolation (TEEs) acts as a secondary layer if a specific hardware enclave is compromised; autonomous node isolation if attestation fails.
7. **Transversal Meta-Layer:** *Sovereignty:* Eliminates "black-box" hardware dependencies. *Sustainability:* Energy-efficient secure silicon designs reducing the carbon footprint of cryptographic operations at the far-edge.

Progression Matrix: Hardware Roots of Trust & EUCC 'High'

Metric	Phase 1: Foundation & Compliance (0-5 Yrs)	Phase 2: Federation & Standardization (5-10 Yrs)	Phase 3: Autonomy & Leadership (10-15 Yrs)
Deployment Phase (ODP)	Fragmented Piloting	Standardized Integration	Ubiquitous Operation
Sovereignty Level (SSL)	High Foreign Dependency	Emerging EU Ecosystem	Full Sovereign Autonomy
Operational State	Cloud-based TPMs common; Edge nodes lack hardware trust; RISC-V pilots begin.	Standardized EU-made secure elements in critical infra; Automated attestation at scale.	Sovereign hardware trust embedded by default in all EU ICT products from IoT to HPC.
Critical Enabler	Cyber Resilience Act (CRA) enforcement	EUCC 'High' certification of EU-based secure elements	Maturity of EU Chips Act sovereign fabrication lines

Priority 2: Post-Quantum Cryptographic Agility for Edge/IoT

1. **Baseline & Target Objective:** *Baseline:* Current IoT and edge infrastructure predominantly relies on classical cryptography (RSA, ECC) that is vulnerable to future quantum computers. There is limited awareness and no systematic migration path for resource-constrained devices. *Target:* A “Quantum-Native” continuum where all critical edge/IoT infrastructure has migrated to hybrid or full Post-Quantum Cryptography (PQC), with automated crypto-agility enabling seamless algorithm updates without hardware replacement.
2. **Primary Drivers:** *Threat:* The **Q-Day** threat—the point when Cryptographically Relevant Quantum Computers (CRQC) become operational. The “Harvest Now, Decrypt Later” strategy where adversaries capture encrypted data today to decrypt once quantum capabilities mature. *Regulatory:* Emerging ENISA guidelines and potential future EU mandates requiring PQC readiness for critical infrastructure under NIS2.
3. **High Impact Activities:** Optimization of NIST-standardized PQC algorithms (e.g., CRYSTALS-Kyber, CRYSTALS-Dilithium) for 8/16-bit microcontrollers; development of crypto-agility middleware enabling over-the-air algorithm updates; creation of automated cryptographic inventory and migration tools for legacy IoT deployments.
4. **Catalysts & Enablers:** **NIST PQC standardization finalization** (2024-2025); **ETSI** quantum-safe cryptography standards; **ENISA PQC guidelines**; **ECCC Strategic Agenda** funding for quantum-resilient infrastructure; integration with EU Chips Act for PQC-native silicon development.
5. **Disruptors & Systemic Risks:** Early arrival of CRQC before migration is complete (Q-Day cliff); performance overhead of PQC algorithms causing unacceptable power consumption on battery-powered IoT devices; fragmented standardization leading to incompatible implementations; insufficient cryptographic inventory of legacy OT/IoT systems preventing complete migration.
6. **Failsafe Mechanisms & Resilience:** Deployment of hybrid cryptographic schemes (classical + PQC) during the transition phase to maintain backward compatibility while providing quantum resistance; implementation of crypto-agility frameworks allowing rapid algorithm switching if vulnerabilities are discovered; diversification of PQC algorithm families to avoid single-point cryptanalytic failures.
7. **Transversal Meta-Layer:** *Sovereignty:* Protection of Europe’s long-term industrial and governmental secrets from future quantum decryption. *Sustainability:* Development of energy-efficient, lightweight PQC implementations reducing the carbon footprint of security (Green IT). *Trust:* Ensuring European citizens that their data remains confidential across decades, not just years.

Progression Matrix: Post-Quantum Cryptographic Agility for Edge/IoT

Metric	Phase 1: Foundation & Compliance (0-5 Yrs)	Phase 2: Federation & Standardization (5-10 Yrs)	Phase 3: Autonomy & Leadership (10-15 Yrs)
Deployment Phase (ODP)	Applied R&D	Fragmented Piloting	Ubiquitous Operation
Sovereignty Level (SSL)	High Foreign Dependency	Emerging EU Ecosystem	Full Sovereign Autonomy
Operational State	NIST algorithm selection and validation; Software-based PQC pilots in cloud environments; Research on lightweight implementations.	Hybrid (classical + PQC) schemes deployed in Industrial IoT and critical edge infrastructure; EU-developed crypto-agility middleware.	Fully quantum-resilient continuum with PQC embedded in all devices; Automated crypto-agility across Cloud-Edge-IoT.
Critical Enabler	ENISA PQC Guidelines and NIST standardization completion	NIST Algorithm Finalization and EU-wide deployment standards	PQC-native RISC-V silicon and ubiquitous crypto-agility frameworks

Priority 3: Security Interoperability via Simpl Middleware

1. **Baseline & Target Objective:** *Baseline:* Current European cloud and edge infrastructure is highly fragmented across proprietary vendor silos, each with incompatible security policies, authentication mechanisms, and data exchange protocols. This fragmentation prevents seamless data flow across Common European Data Spaces and creates vendor lock-in. *Target:* A unified European “Trust Fabric” enabling secure, federated interoperability across heterogeneous cloud and edge providers through mandatory integration with the **Simpl** open-source smart middleware stack, with standardized security Minimal Interoperability Mechanisms (MIMs).
2. **Primary Drivers:** *Regulatory:* **Data Act** mandates requiring seamless data portability and interoperability; requirements for Common European Data Spaces (Health, Manufacturing, Agri-Food, etc.) to operate across multiple providers; NIS2 requirements for coordinated incident response across federated infrastructure. *Market:* Breaking vendor lock-in to enable European SMEs to compete; enabling cross-border data flows while maintaining sovereignty.
3. **High Impact Activities:** Implementation and standardization of security-specific **Minimal Interoperability Mechanisms (MIMs)** within the Simpl stack; development of automated policy translation engines converting provider-specific security policies into a common, machine-readable format (e.g., XACML, OSCAL); creation of federated identity bridges connecting disparate IAM systems using verifiable credentials; deployment of cross-domain security orchestration enabling unified threat response.
4. **Catalysts & Enablers:** **Digital Europe Programme** funding for Simpl development and pilot deployments; **Simpl open-source stack** community and governance; alignment with **Gaia-X** federation services and trust frameworks; ECCC

support for security MIM standardization; CEN/CENELEC standardization of interoperability protocols.

5. **Disruptors & Systemic Risks:** Vendor resistance to open standards threatening adoption; weakest-link security failures in federated systems where one provider’s compromise cascades across the entire fabric; complexity of policy translation creating semantic gaps or enforcement inconsistencies; fragmented national implementations creating “data islands” rather than true European interoperability.
6. **Failsafe Mechanisms & Resilience:** Capability to autonomously “unfederate” or isolate nodes that fail to meet the shared security baseline, preventing compromise propagation; implementation of zero-trust architectures where federation does not imply implicit trust; continuous compliance monitoring with automated ejection of non-compliant providers; cryptographic attestation of policy enforcement ensuring verifiable security guarantees.
7. **Transversal Meta-Layer:** *Sovereignty:* Breaking dependence on non-EU hyper-scaler lock-in by enabling European providers to compete on equal footing. *Trust:* Fostering pan-European cross-border trust through transparent, verifiable security mechanisms. *Economic Viability:* Enabling SMEs to participate in Data Spaces without massive security infrastructure investments through standardized, reusable security components.

Progression Matrix: Security Interoperability via Simpl Middleware

Metric	Phase 1: Foundation & Compliance (0-5 Yrs)	Phase 2: Federation & Standardization (5-10 Yrs)	Phase 3: Autonomy & Leadership (10-15 Yrs)
Deployment Phase (ODP)	Fragmented Piloting	Standardized Integration	Ubiquitous Operation
Sovereignty Level (SSL)	Emerging EU Ecosystem	Full Sovereign Autonomy	Full Sovereign Autonomy
Operational State	Initial Simpl security pilots in select Data Spaces; Manual policy coordination; Proof-of-concept MIM implementations.	Automated policy translation via standardized MIMs; Simpl-based security deployed in majority of Common European Data Spaces; Cross-border federated SOC operations.	Fully automated, provider-agnostic European trust fabric; Real-time security orchestration across the continuum; Zero-trust federation by default.
Critical Enabler	Simpl-Open official launch and initial MIM definitions	Security MIMs adoption as CEN/CENELEC standards	EU Cloud Rulebook maturity and Data Act full enforcement

12 Executive Summary & Strategic Recommendations

Executive Summary

This cybersecurity and cyber-resilience strategy defines a comprehensive 15-year trajectory to secure the European Cloud-Edge-IoT Continuum against emerging threats while

establishing digital sovereignty. The strategy moves beyond traditional perimeter defenses toward autonomous, federated, and certified Security-by-Design architectures.

By utilizing a **4-Filter Prioritization Framework** (Regulatory Urgency, Foundational Dependency, Sovereignty Cliff, and Market Feasibility), we identify three critical path priorities for Phase 1 (2025-2030): **Hardware Roots of Trust & EUCC 'High' Certification**, **Post-Quantum Cryptographic Agility for Edge/IoT**, and **Security Interoperability via Simpl Middleware**. These priorities ensure that foundational infrastructure is hardened against both physical and quantum threats while enabling secure, sovereign cross-border data flows.

The roadmap employs a dual-metric progression framework: **Operational Deployment Phase (ODP)** tracking market integration, and **Strategic Sovereignty Level (SSL)** measuring European independence from foreign technology dependencies. This enables objective tracking of capability maturity across three distinct phases: Foundation & Compliance (0-5 years), Federation & Standardization (5-10 years), and Autonomy & Leadership (10-15 years).

Strategic Recommendations for Phase 1 (2025–2030)

To operationalize this strategy, the following actions are recommended for immediate European R&I funding and policy alignment:

- **Accelerate Sovereign Secure Silicon Development:** Immediate Horizon Europe and EU Chips Act funding for open-source RISC-V secure elements and hardware roots of trust engineered to meet **EUCC 'High' assurance levels**. Establish pilot fabrication lines dedicated to security-critical components to reduce 100% foreign dependency on TPMs and secure enclaves.
- **Operationalize Crypto-Agility for the Edge/IoT:** Develop and standardize automated crypto-agility middleware frameworks that allow resource-constrained IoT devices to upgrade to **Post-Quantum Cryptography (PQC)** without requiring hardware replacement. Prioritize ENISA guidelines alignment and ECCC funding for lightweight PQC implementations targeting 8/16-bit microcontrollers.
- **Enforce Security-by-Design via Simpl Integration:** Mandate standardized security **Minimal Interoperability Mechanisms (MIMs)** across all Common European Data Spaces to ensure security policies are provider-agnostic, cryptographically verifiable, and eliminate vendor lock-in. Align Digital Europe Programme funding with Simpl stack security module development.
- **Harmonize Continuous Certification Lifecycles:** Transition from static, point-in-time PDF certificates to **Continuous Automated Assurance** frameworks using telemetry-driven compliance verification. Develop real-time certification tracking standards for both cloud services (**EUCCS**) and physical edge products (**EUCC**), ensuring autonomous compliance maintenance throughout operational lifecycles.
- **Establish Federated Cyber Resilience Capabilities:** Invest in cross-border SOC federation infrastructure and legally protected threat intelligence sharing platforms to overcome institutional trust barriers between Member States. Align with CONCORDIA project outcomes and ECCC Strategic Agenda priorities for collaborative defense.

- **Synchronize Regulatory Enforcement Timelines:** Ensure coordinated enforcement of NIS2, Cyber Resilience Act (CRA), and AI Act requirements by 2026-2027 to create unified market incentives for security investment. Provide targeted support for SMEs to meet compliance requirements without creating insurmountable barriers to market entry.

Success Metrics for 2030

By the end of Phase 1, Europe should achieve:

- **Hardware Sovereignty:** At least 30% of critical infrastructure edge nodes utilizing EU-designed, EUCC 'High' certified secure elements.
- **Quantum Readiness:** 100% of new critical IoT deployments supporting hybrid PQC schemes; complete cryptographic inventory of legacy systems.
- **Federated Trust:** Majority of Common European Data Spaces operating on Simpl-based security interoperability with standardized MIMs.
- **Certification Maturity:** Operational continuous certification frameworks for at least two major Data Spaces with real-time EUCS/EUCC tracking.

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